

JAIDEV EDUCATION SOCIETY'S
J D COLLEGE OF ENGINEERING & MANAGEMENT, NAGPUR



Department of Computer Science & Engineering

SESSION: 2025-2026 5th SEM- CSE

Subject: Blockchain Technology

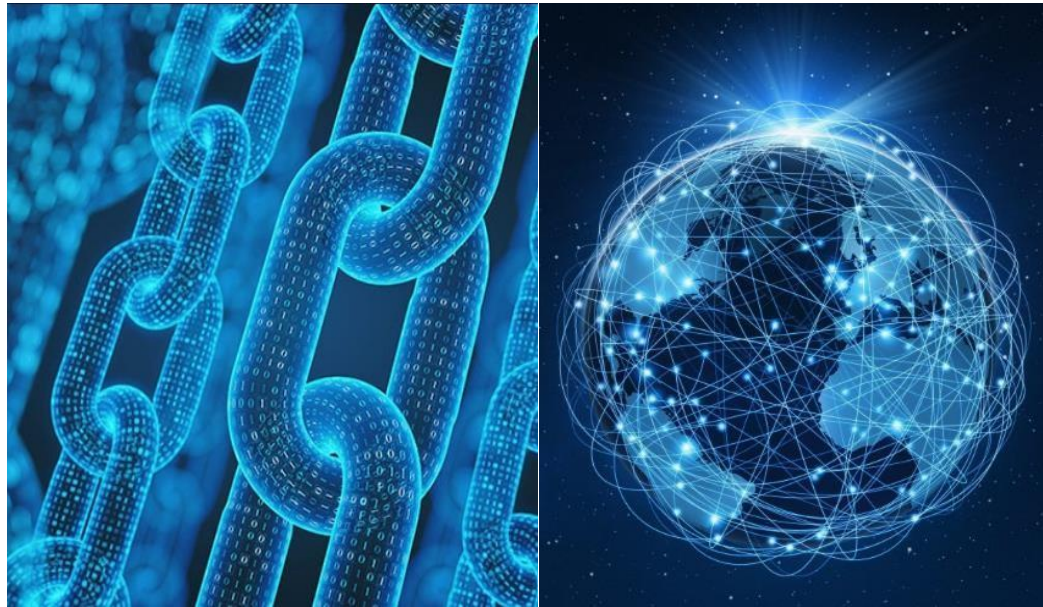
UNIT-I Notes

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Block chain

- The technology behind crypto currencies.
- Analogous to the internet



When we
have internet

Why BlockChain?

Problems which Internet failed to solve??

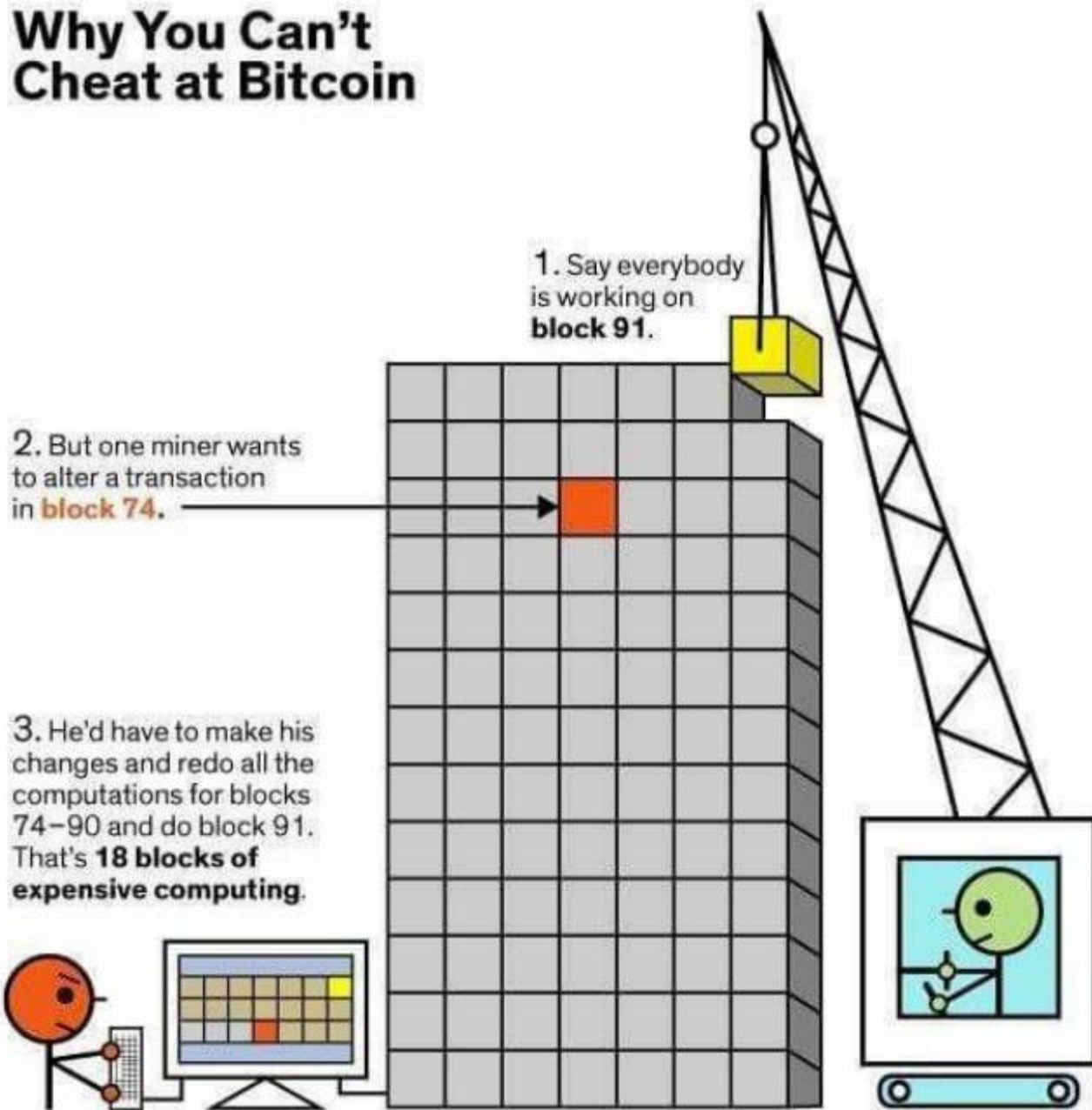
Trust



Intermediary



Why You Can't Cheat at Bitcoin



4. What's worse, he'd have to do it all **before** everybody else in the Bitcoin network finished **just the one block (number 91)** that they're working on.

Block chain is a revolutionary technology that functions as a shared, immutable digital ledger. The name "blockchain" comes from its structure data is organized in blocks, with each new block linked to the one before it, forming a continuous chain.

Each block contains crucial data, such as a list of transactions, a timestamp, and a unique identifier called a cryptographic hash. This hash is generated from the block's contents and the hash of the previous block, ensuring that each block is tightly connected to the one before it.

Block chain is decentralized and distributed, meaning no single authority controls it. Instead, multiple computers (nodes) on a network each have a copy of the blockchain, keeping the ledger synchronized. This setup ensures that once data, like a transaction, is recorded and confirmed, it becomes immutable almost impossible to alter or delete

Building trust with Blockchain: Blockchain enhances trust across a business network. It's not that you can't trust those who you conduct business with it's that you don't need to when operating on a Blockchain network. Blockchain builds trust through the following five attributes:

- **Distributed:** The distributed ledger is shared and updated with every incoming transaction among the nodes connected to the Blockchain. All this is done in real time as there is no central server controlling the data.
- **Secure:** There is no unauthorized access to Blockchain made possible through Permissions and Cryptography.
- **Transparent:** Because every node or participant in Blockchain has a copy of the Blockchain data, they have access to all transaction data. They themselves can verify the identities without the need for mediators.

- **Consensus-based:** All relevant network participants must agree that a transaction is valid. This is achieved through the use of consensus algorithms.
- **Flexible:** Smart Contracts which are executed based on certain conditions can be written into the platform. Blockchain Networks can evolve in pace with business processes.

Key Components of Blockchain

Blockchain technology relies on a few fundamental components and concepts that work together to achieve its functionality:

- **Distributed Ledger:** Blockchain is a shared record system spread across many computers, with each participant having a copy. Once data is added, it can't be changed or deleted, ensuring no single point of failure or control.
- **Blocks:** Data is stored in **blocks**, each containing a set of transactions, a timestamp, and a reference (hash) to the previous block. This creates a secure, linked chain of blocks, where any change would disrupt the chain.
- **Nodes (Peer-to-Peer Network):** **Nodes** are the devices in the network that store the blockchain and validate new transactions. They communicate directly, ensuring the blockchain remains decentralized and operates without a central authority.
- **Cryptography (Hashes & Signatures):** Blockchain uses **cryptography** to secure data. Each block's unique **hash** acts as a tamper-proof seal, and public/private key pairs verify the authenticity of transactions, ensuring data integrity and privacy.

- **Consensus Mechanism:** Nodes agree on the blockchain's state using consensus algorithms (e.g., **Proof of Work** or **Proof of Stake**). This prevents fraud by ensuring that the majority of the network agrees on each new block.
- **Smart Contracts:** **Smart contracts** are automated scripts on the blockchain that execute actions when specific conditions are met. They help automate complex processes like payments and agreements, without the need for intermediaries, providing transparency and security.

History and Evolution of Blockchain

In 1982 Cryptographer David Chaum proposed a protocol, that resembles blockchain elements, aiming to secure information among network participants. In the early 1990s, Stuart Haber and W. Scott Stornetta created the first prototype of a cryptographically secured chain of blocks, using hashing to time-stamp documents to prevent tampering. This led to the development of Merkle trees in 1992, improving efficiency.

Bitcoin was created in 2008 when an unknown entity published a white paper, using the name Satoshi Nakamoto. Bitcoin introduced a decentralized, public blockchain that solved the double-spending problem in digital money, allowing secure transactions without relying on banks or governments.

Advantages of Blockchain Technology:

1. **Decentralization:** The decentralized nature of blockchain technology eliminates the need for intermediaries, reducing costs and increasing transparency.
2. **Security:** Transactions on a blockchain are secured through cryptography, making them virtually immune to hacking and fraud.

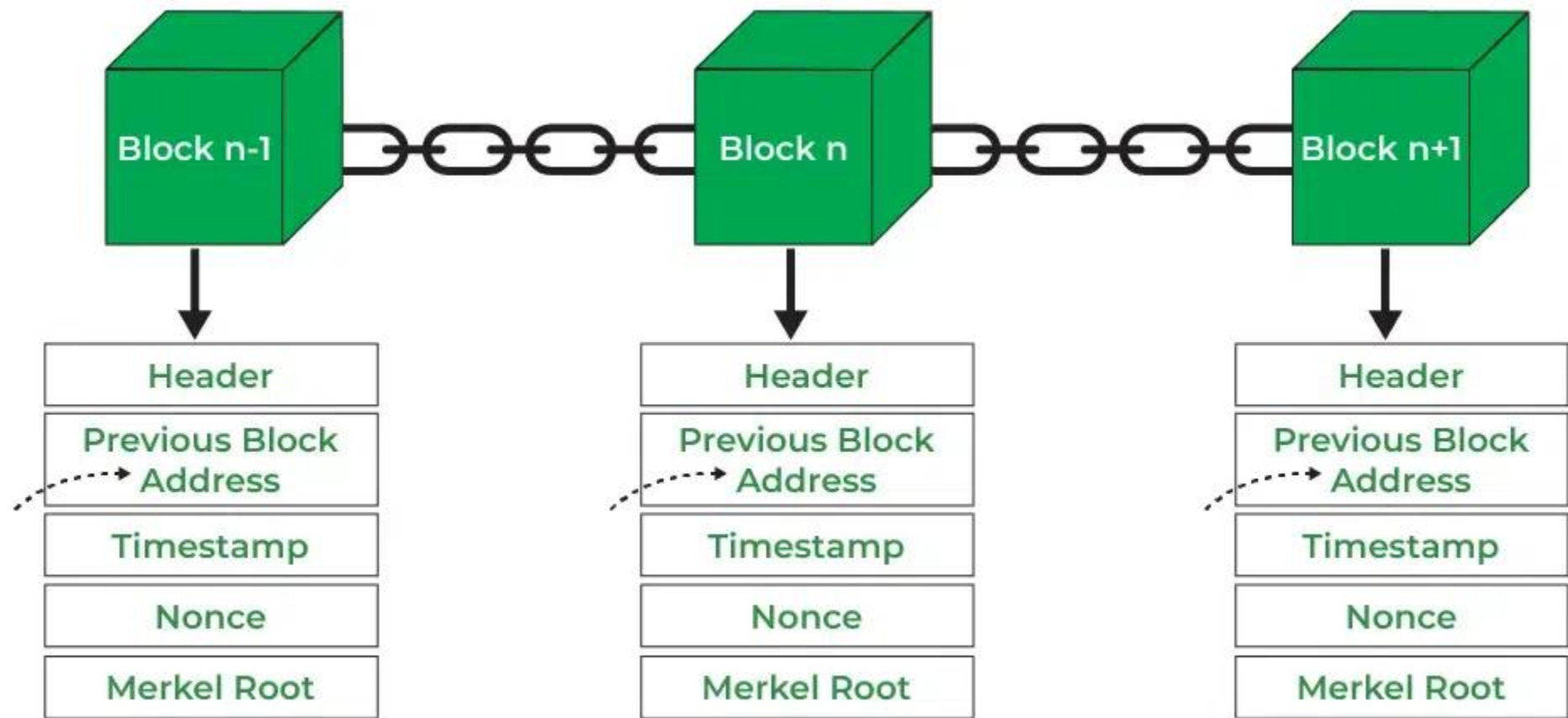
3. **Transparency:** Blockchain technology allows all parties in a transaction to have access to the same information, increasing transparency and reducing the potential for disputes.
4. **Efficiency:** Transactions on a blockchain can be processed quickly and efficiently, reducing the time and cost associated with traditional transactions.
5. **Trust:** The transparent and secure nature of blockchain technology can help to build trust between parties in a transaction.

Disadvantages of Blockchain Technology:

1. **Scalability:** The decentralized nature of blockchain technology can make it difficult to scale for large-scale applications.
2. **Energy Consumption:** The process of mining blockchain transactions requires significant amounts of computing power, which can lead to high energy consumption and environmental concerns.
3. **Adoption:** While the potential applications of blockchain technology are vast, adoption has been slow due to the technical complexity and lack of understanding of the technology.
4. **Regulation:** The regulatory framework around blockchain technology is still in its early stages, which can create uncertainty for businesses and investors.
5. **Lack of Standards:** The lack of standardized protocols and technologies can make it difficult for businesses to integrate blockchain technology into their existing systems.

Data Storage and Management

1. **Header:** It is used to identify the particular block in the entire blockchain. It handles all blocks in the blockchain. A block header is hashed periodically by miners by changing the nonce value as part of normal mining activity, also Three sets of block metadata are contained in the block header.
2. **Previous Block Address/ Hash:** It is used to connect the $i+1$ th block to the i th block using the hash. In short, it is a reference to the hash of the previous (parent) block in the chain.
3. **Timestamp:** It is a system that verifies the data into the block and assigns a time or date of creation for digital documents. The timestamp is a string of characters that uniquely identifies the document or event and indicates when it was created.
4. **Nonce:** A nonce number which used only once. It is a central part of the proof of work in the block. It is compared to the live target if it is smaller or equal to the current target. People who mine, test, and eliminate many Nonce per second until they find that Valuable Nonce is valid.
5. **Merkel Root:** It is a type of data structure frame of different blocks of data. A [Merkle Tree](#) stores all the transactions in a block by producing a digital fingerprint of the entire transaction. It allows the users to verify whether a transaction can be included in a block or not.

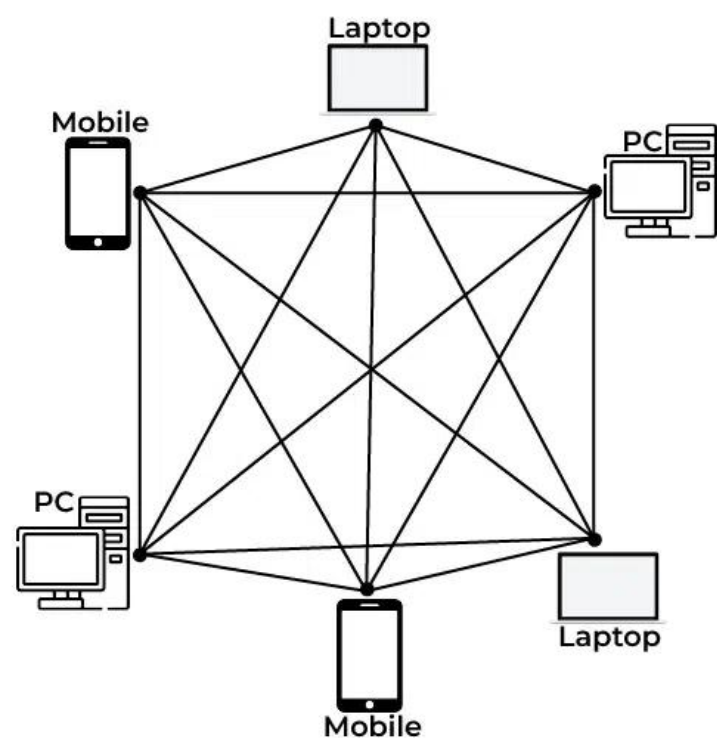


Types of Blockchain

1. Public Blockchain

A public blockchain is a concept where anyone is free to join and take part in the core activities of the blockchain network. Anyone can read, write, and audit the ongoing activities on a public blockchain network, which helps to achieve the self-determining, decentralized nature often authorized when blockchain is discussed. Data on a public blockchain is secure as it is not possible to modify once they are validated.

The public blockchain is fully decentralized, it has access and control over the ledger, and its data is not restricted to persons, is always available and the central authority manages all the blocks in the chain. There is publicly running all operations. Due to no one handling it singly then there is no need to get permission to access the public blockchain. Anyone can set his/her node or block in the network/ chain. After a node or a block settles in the chain of the blocks, all the blocks are connected like peer-to-peer connections. If someone tries to attack the block then it forms a copy of that data and it is accessible only by the original author of the block.



Public Blockchain

Advantages:

1. **Decentralization:** High level of decentralization, which reduces the risk of single points of failure and increases security.
2. **Transparency:** All transactions are visible to anyone, enhancing transparency and trust.
3. **Immutability:** Once data is recorded, it cannot be altered or deleted, providing a permanent record.

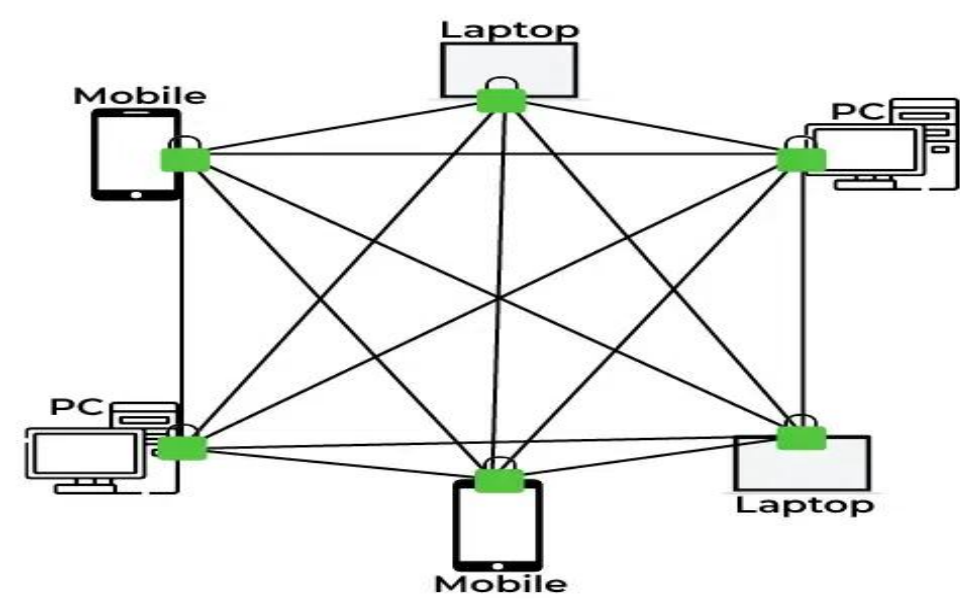
Disadvantages:

1. **Scalability Issues:** Public blockchains often face scalability problems, with limited transaction throughput and slower processing times.

2. **Energy Consumption:** Some consensus mechanisms, like Proof of Work (PoW), require significant computational power and energy.
3. **Privacy Concerns:** Public visibility of transactions may lead to privacy issues, as sensitive data can be exposed.

2. Private Blockchain

Miners need permission to access a [private blockchain](#). It works based on permissions and controls, which limit participation in the network. Only the entities participating in a transaction will know about it and the other stakeholders not be able to access it. It works based on permissions due to this it is also called a permission-based blockchain. Private blockchains are not like public blockchains it is managed by the entity that owns the network. A trusted person is in charge of the running of the blockchain it will control who can access the private blockchain and also control the access rights of the private chain network. There may be a possibility of some restrictions while accessing the network of the private blockchain.



Private Blockchain

Advantages:

1. **Performance and Speed:** Faster transaction processing and higher throughput compared to public blockchains due to fewer nodes and reduced consensus requirements.
2. **Privacy:** Transactions and data are visible only to authorized participants, enhancing privacy.
3. **Control:** Centralized control allows for easier governance and compliance with regulations.

Disadvantages:

1. **Centralization:** Less decentralized than public blockchains, which can introduce single points of failure and reduce the security benefits.
2. **Trust:** Requires participants to trust the central authority or consortium managing the blockchain.

3. **Limited Transparency:** Reduced transparency can make it harder for external auditors to verify data.

3. Consortium Blockchain

A consortium blockchain is a concept where it is permissioned by the government and a group of organizations, not by one person like a private blockchain. Consortium blockchains are more decentralized than private blockchains, due to being more decentralized it increases the privacy and security of the blocks. Those like private blockchains connected with government organizations' block networks. Consortium blockchains lie between public and private blockchains. They are designed by organizations and no one person outside of the organizations can gain access. In Consortium blockchains all companies in between organizations collaborate equally. They do not give access from outside of the organizations/consortium network.

Advantages:

1. **Efficiency:** Better performance and efficiency than public blockchains due to fewer nodes and optimized consensus mechanisms.
2. **Shared Control:** Governance is shared among consortium members, which can enhance trust and cooperation.
3. **Privacy and Security:** Improved privacy and security compared to public blockchains, as access is restricted.

Disadvantages:

1. **Complex Governance:** Decision-making can be complex due to multiple stakeholders with potentially conflicting interests.

2. **Less Decentralization:** While more decentralized than private blockchains, consortium blockchains still have a limited number of participants, which can reduce some decentralization benefits.
3. **Interoperability:** Challenges can arise when integrating with other blockchain networks or systems.

4. Hybrid Blockchain

Hybrid blockchains combine elements of both public and private blockchains. They aim to offer the benefits of both types, allowing for controlled access and transparency. Examples include Dragonchain and IBM's Food Trust.

Advantages:

1. **Flexibility:** Offers the ability to balance transparency and privacy based on the needs of the organization or project.
2. **Scalability and Performance:** Can be designed to optimize performance and scalability while maintaining some degree of transparency.
3. **Customizable Access:** Allows organizations to control who can access certain data while making some data available to the public.

Disadvantages:

1. **Complexity:** Implementation can be more complex due to the need to manage different access levels and integrations.
2. **Governance Challenges:** Balancing governance between public and private aspects can be challenging.

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3. **Potential for Confusion:** This may require lead to confusion among users and stakeholders about the nature and scope of access and transparency.

Different Layers of Blockchain

In the domain of blockchain technology, an ecosystem is a collection of components that work together to produce a unique environment. Blockchain-based ecosystems have a collection of distributed nodes where immutable transactions are repeated. The blockchain (or digital ledger) technology is built upon a layered architecture as shown in the below figure.

According to some blockchain professionals, there are five layers of blockchain technology:

1. **Hardware/Infrastructure layer.**
2. **Data layer.**
3. **Network layer.**
4. **Consensus layer.**
5. **Application and Presentation layer.**

Some blockchain experts believe that blockchain has 7 layers:

1. **Hardware/Infrastructure layer.**
2. **Data layer.**
3. **Network layer.**
4. **Consensus layer.**

- 5. **Incentive layer.**
- 6. **Contract layer.**
- 7. **Application and Presentation layer.**

The 7 Blockchain layers are explained below:

1. Hardware or Infrastructure Layer

- Hardware or infrastructure makes up the Blockchain's top layer. On a server housed in a data center, the content for the Blockchain is hosted.
- Client-server architecture, as used in web browsing and other applications, refers to the process where clients request data or content from application servers.
- Blockchain technology typically uses a Peer-to-Peer (P2P) network of computers to calculate, validate, and record transactions in an ordered format in a shared ledger.
- A distributed database is the end result, and it keeps track of all the data, transactions, and other pertinent information.
- Nodes are the computers that make up a peer-to-peer network. The Blockchain network's nodes are responsible for verifying transactions, grouping them into blocks, broadcasting them to the network, and so forth.
- In the event that all parties agree, the nodes update their local copies of the ledger and commit the block to the Blockchain network. A device is referred to and utilized as a node after it connects to a blockchain network.

2. Data Layer

- The linked list used to represent a blockchain's data structures has two main components: pointers and a linked list of blocks that arrange the transactions.
- A linked list is a collection of chained blocks where each block is filled with data and pointers to the block before it. Pointers are variables that refer to the location of other variables.
- A Merkle tree is a binary tree of hashes that holds the hash of the Merkle root along with other information including the hash of the previous block, the date, the nonce, the block version number, and the current difficulty goal.
- Blockchain technology is secured, has integrity, and is irrefutable due to a Merkle tree.
- A hash tree, also known as a Merkle tree, is a tree in which each leaf node is labeled with the cryptographic hash of a data block and each non-leaf node is labeled with the labels of all of its child nodes.
- In order to guarantee the security and integrity of the data stored there, transactions are digitally signed on the blockchain.
- Anybody with the public key can validate the signer of a transaction that has been signed using a private key.
- Because the encrypted data is also signed, the digital signature ensures data integrity and verifies information tampering.
- Data encryption prevents the detection of the information. The sender's or owner's identity is also protected by a digital signature.
- The structure of a block on a blockchain is determined by the data layer.

3. Network Layer

- The network layer is also known as the peer-to-peer (P2P) layer. In addition to being known as the propagation layer, it is in charge of inter-node communication.
- Transactions, block propagation, and discovery are handled by the network layer.
- The network layer makes sure that nodes are able to communicate, synchronize, and propagate with one another in order to keep the blockchain network's present state valid.
- Peer-to-peer networks are types of computer networks where nodes are dispersed and share workloads in order to accomplish a common purpose. Nodes also carry out blockchain transactions.
- The terms "full node" and "light node" refer to two different types of nodes. Ensured by full nodes include mining, consensus rule enforcement, and transaction validation and verification. Light nodes, however, can send transactions and merely store the Blockchain header.

4. Consensus Layer

- Blockchain platforms cannot function without the consensus layer. Whether using Ethereum, Hyperledger or another blockchain, the consensus layer is the most important and fundamental layer.
- The consensus layer is in charge of validating the blocks, putting them in the proper sequence, and making sure everyone is in agreement.
- The distributed peer-to-peer network's consensus layer establishes a certain set of agreements between nodes.
- Power remains distributed and decentralized due to the consensus layer.

5. Incentive Layer

This stack's optional incentive layer is the fourth tier. This layer handles how network nodes are compensated for the effort they put forth to establish consensus in terms of rewards. Depending on the consensus process being used, this layer may or may not be implemented.

- This layer defines the minimum amount of transaction fees needed to perform actions on the blockchain.
- This determines the variant types of incentives available on the network.

6. Contract Layer

The information in the contracts layer, which is right next to the application layer and specifies how a service will operate and what kind of information will be made accessible, is similar to that in a real-world contract. In essence, there are four types of contracts, which are briefly discussed below:

- **Service contract:** The endpoint's offerings and the protocols to be used in the communication process are described in this contract for the benefit of the client as well as the general public.
- **Data contract:** A data contract lays out the terms of the data that a service will exchange. The data contract must be accepted by the service as well as the client.
- **Message contract:** An agreement over a message governs a data contract. Its main function is to customize the SOAP message parameters' type formatting. The SOAP format is used by WCF for communication purposes, it should be noted. Simple Object Access Protocol is the abbreviation for this.

- **Policy and Binding:** There are a few prerequisites that must be met in order to communicate with a service, and these prerequisites are set forth in the policy and legally binding agreements. This contract must be adhered to by the client.
- The application layer is subdivided into the execution layer and the application layer.
- The programs that end users utilize to engage with the blockchain network are included in the application layer.
- Scripts, APIs, user interfaces, and frameworks make up the application layer, together with smart contracts, chaincode, and decentralized applications (dApps). The blockchain network serves as the back-end system for various applications, which are connected to it through APIs.
- Chain code, smart contracts, and underlying rules make up the execution layer, which is a sublayer.
- However, the transaction is validated and carried out at the semantic layer before moving from the application layer to the execution layer.
- Applications transmit instructions to the execution layer, which carries out transaction processing and maintains the Blockchain's deterministic nature.

Smart Contracts in Blockchain

A Smart Contract (or cryptocontract) is a computer program that directly and automatically controls the transfer of digital assets between the parties under certain conditions. A smart contract works in the same way as a traditional contract while also automatically enforcing the contract. Smart contracts are programs that execute exactly as they are set up(coded, programmed) by their creators. Just like a traditional contract is enforceable by law, smart contracts are enforceable by code.

- The [bitcoin](#) network was the first to use some sort of smart contract by using them to transfer value from one person to another.
- The smart contract involved employs basic conditions like checking if the amount of value to transfer is actually available in the sender account.
- Later, the [Ethereum](#) platform emerged which was considered more powerful, precisely because the developers/programmers could make custom contracts in a Turing-complete language.
- It is to be noted that the contracts written in the case of the bitcoin network were written in a Turing-incomplete language, restricting the potential of smart contracts implementation in the bitcoin network.
- There are some common smart contract platforms like Ethereum, Solana, Polkadot, [Hyperledger fabric](#), etc.

Capabilities of Smart Contracts

1. **Accuracy:** Smart contracts are accurate to the limit a programmer has accurately coded them for execution.
2. **Automation:** Smart contracts can automate the tasks/ processes that are done manually.
3. **Speed:** Smart contracts use software code to automate tasks, thereby reducing the time it takes to maneuver through all the human interaction-related processes. Because everything is coded, the time taken to do all the work is the time taken for the code in the smart contract to execute.
4. **Backup:** Every node in the [blockchain](#) maintains the shared ledger, providing probably the best backup facility.
5. **Security:** [Cryptography](#) can make sure that the assets are safe and sound. Even if someone breaks the encryption, the hacker will have to modify all the blocks that come after the block which has been modified. Please note that this is a highly difficult and computation-intensive task and is practically impossible for a small or medium-sized organization to do.
6. **Savings:** Smart contracts save money as they eliminate the presence of intermediaries in the process. Also, the money spent on the paperwork is minimal to zero.
7. **Manages information:** Smart contract manages users' agreement, and stores information about an application like domain registration, membership records, etc.
8. **Multi-signature accounts:** Smart contracts support multi-signature accounts to distribute funds as soon as all the parties involved confirm the agreement.

How Do Smart Contracts Work?

A smart contract is just a digital contract with the security coding of the blockchain.

- It has details and permissions written in code that require an exact sequence of events to take place to trigger the agreement of the terms mentioned in the smart contract.
- It can also include the time constraints that can introduce deadlines in the contract.
- Every smart contract has its address in the blockchain. The contract can be interacted with by using its address presuming the contract has been broadcasted on the network.

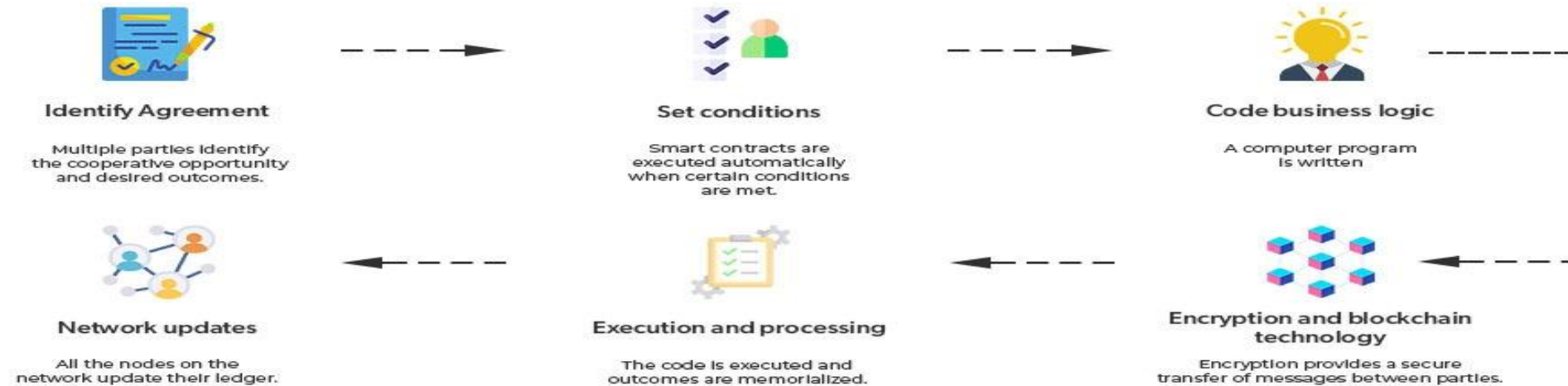
The idea behind smart contracts is pretty simple. They are executed on a basis of simple logic, IF-THEN for example:

- **IF** you send object A, **THEN** the sum (of money, in cryptocurrency) will be transferred to you.
- **IF** you transfer a certain amount of digital assets (cryptocurrency, for example, ether, bitcoin), **THEN** the A object will be transferred to you.
- **IF** I finish the work, **THEN** the digital assets mentioned in the contract will be transferred to me.

Note: The WHEN constraint can be added to include the time factor in the smart contracts. It can be seen that these smart contracts help set conditions that have to be fulfilled for the terms of the contract agreement to be executed. There is no limit on how much IF or THEN you can include in your intelligent contract.

Smart Contract Working

How does a Smart Contract Work?



- **Identify Agreement:** Multiple parties identify the cooperative opportunity and desired outcomes and agreements could include business processes, asset swaps, etc.
- **Set conditions:** Smart contracts could be initiated by parties themselves or when certain conditions are met like financial market indices, events like GPS locations, etc.
- **Code business logic:** A computer program is written that will be executed automatically when the conditional parameters are met.
- **Encryption and blockchain technology:** Encryption provides secure authentication and transfer of messages between parties relating to smart contracts.

- **Execution and processing:** In blockchain iteration, whenever consensus is reached between the parties regarding authentication and verification then the code is executed and the outcomes are memorialized for compliance and verification.
- **Network updates:** After smart contracts are executed, all the nodes on the network update their ledger to reflect the new state. Once the record is posted and verified on the blockchain network, it cannot be modified, it is in append mode only.

• **How does the Blockchain Work?**

A blockchain is a distributed database that stores information electronically in a digital format and is shared among the nodes of a computer network. A typical difference between a blockchain and a database is how data is structured. A blockchain is a shared, immutable ledger as the name suggests structures data into chunks or blocks, and a database structures data into tables. A blockchain is a chain of blocks. Once a block is filled with data it is chained to the previous blocks. Different types of information can be stored on the blockchain network but the most important is transactions.

How Does a Blockchain Work?
The transaction process in a blockchain can be summarized as follows:

- 1. Facilitating a transaction:** A new transaction enters the blockchain network. All the information that needs to be transmitted is doubly encrypted using public and private keys.
- 2. Verification of transaction:** The transaction is then transmitted to the network of peer-to-peer computers distributed across the world. All the nodes on the network will check for the validity of the transaction like if a sufficient balance is available for carrying out the transaction.

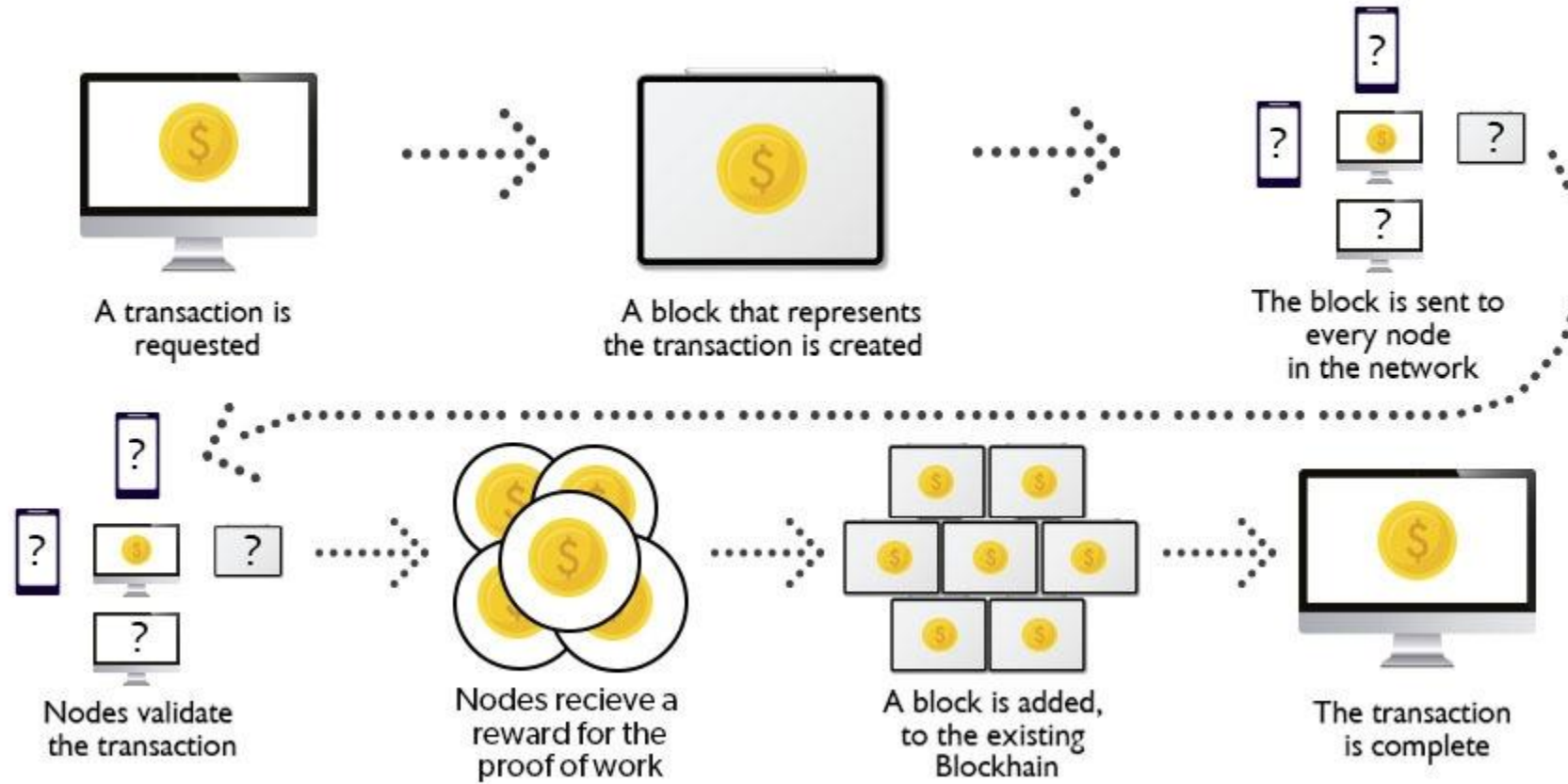
3. Formation of a new block: In a typical blockchain network there are many nodes and many transactions get verified at a time. Once the transaction is verified and declared a legitimate transaction, it will be added to the mempool. All the verified transactions at a particular node form a mempool and such multiple mempools form a block.

4. Consensus Algorithm: The nodes that form a block will try to add the block to the blockchain network to make it permanent. But if every node is allowed to add blocks in this manner then it will disrupt the working of the blockchain network. To solve this problem, the nodes use a consensus mechanism to ensure that every new block that is added to the Blockchain is the only version of the truth that is agreed upon by all the nodes in the Blockchain, and only a valid block is securely attached to the blockchain. The node that is selected to add a block to the blockchain will get a reward and hence we call them "miners". The consensus algorithm creates a hash code for that block which is required to add the block to the blockchain.

5. Addition of the new block to the blockchain: After the newly created block has got its hash value and is authenticated, now it is ready to be added to the blockchain. In every block, there is a hash value of the previous block and that is how the blocks are cryptographically linked to each other to form a blockchain. A new block gets added to the open end of the blockchain.

6. Transaction complete: As soon as the block is added to the blockchain the transaction is completed and the details of this transaction are permanently stored in the blockchain. Anyone can fetch the details of the transaction and confirm the transaction.

How Blockchain Works?



Let's understand this working of blockchain with the help of an example:

Let's say Jack and Phil are two nodes on the [bitcoin](#) blockchain network who wants to carry out a transaction between them.

Step 1: Facilitating the transaction: Jack wants to send 20 BTC to Phil via the Blockchain network.

Step 2: Verification of transaction: The message for verification will be sent to all the nodes on the network. All the nodes will check the important parameters related to the transaction like Does Jack has sufficient balance i.e. at least 20BTC to perform the transaction. Is Jack a registered node? Is Phil a registered node? After checking the parameters the transaction is verified.

Step 3: Formation of a new block: A number of verified transactions stack up in mempools and get stored in a block. This verified transaction will also get stored in a block.

Step 4: Consensus algorithm: Since here we are talking about bitcoins so the [Proof-of-Work consensus algorithm](#) will be used for block verification. In proof-of-work, the system assigns the target hash value to a node, and according to this, it must come up with a hash for the new block. The node has to calculate the hash value for the new block that is less than the target value. If two or more miners mine the same block at the same time, the block with more difficulty is selected. The others are known as stale blocks. Mining usually rewards miners with blockchain currency. In this case, the blockchain currency is bitcoin.

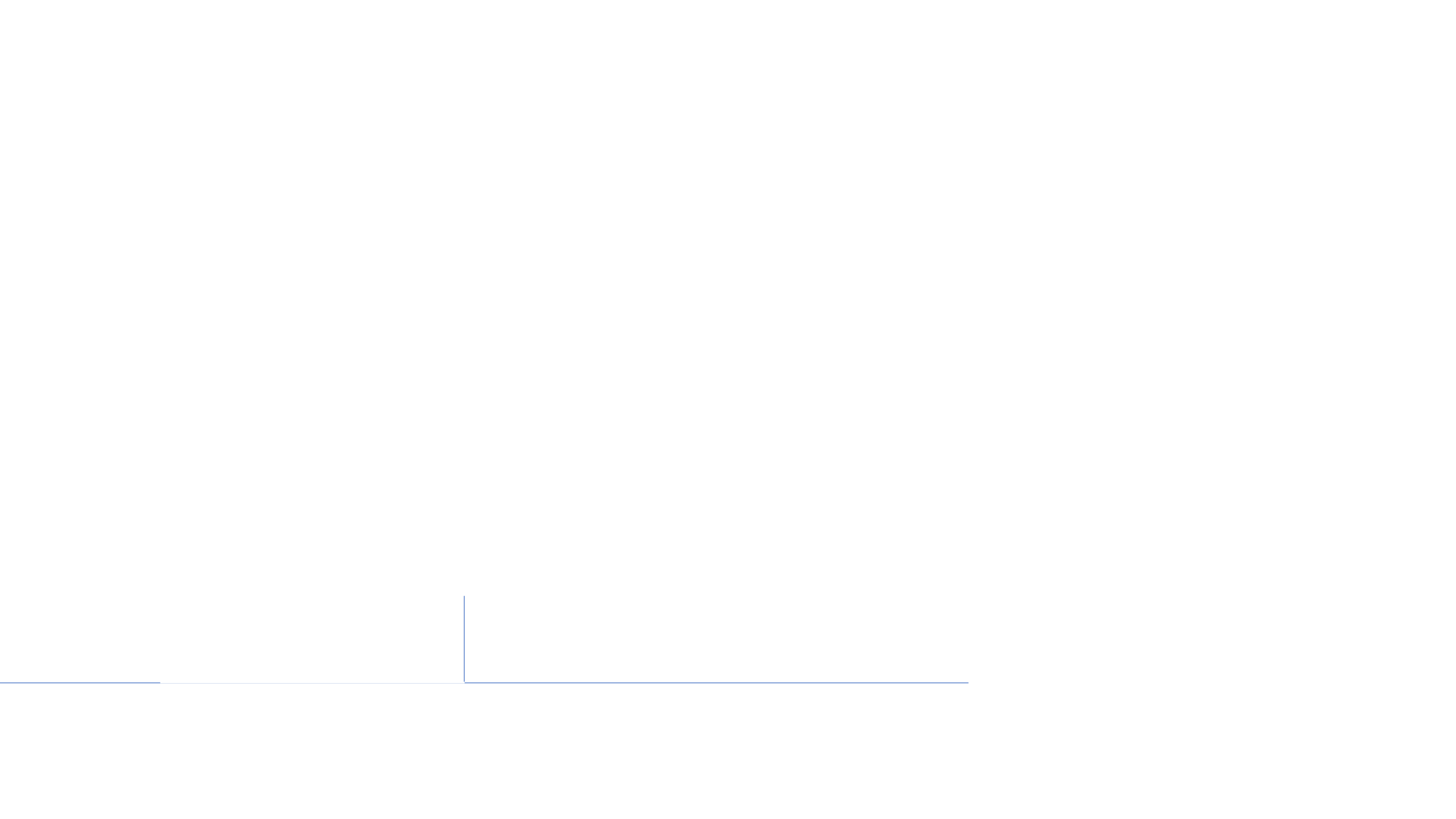
Step 5: Addition of the new block in the blockchain: After the newly created block has got the hash value and authentication through proof-of-work only then it will be added to the network and the transaction will mark as complete. Phil will receive 20 BTC from Jack. The new block will be linked to the open end of the blockchain.

Step 6: Transaction complete: As soon as the block is added to the blockchain, the transaction will take place and 20 BTCs will get transferred from Jack's wallet to Phil's wallet. The details of the transaction are permanently secured on the blockchain.

Anyone on the network can fetch the information and confirm the transaction. This will help to keep track of all the transactions and to verify whether any user is trying to double spend. For example, if Jack tries to carry out a transaction in the future, the rest of the nodes can check Jack's past transaction records to check whether Jack has enough balance to carry out the current transaction. If there is enough balance then the transaction will be approved.

Core Blockchain component

- User or computer within the blockchain architecture (each has an independent copy of the whole blockchain ledger)
- Smallest building block of a blockchain system (records, information, etc.) that serves as the purpose of blockchain
- a data structure used for keeping a set of transactions which is distributed to all nodes in the network
- a sequence of blocks in a specific order
- specific nodes which perform the block verification process before adding anything to the blockchain structure



Applications of Blockchain Technology



Cryptocurrencies



Blockchains for
everything else



Records Management



Banking and Finance



Property



**Online Content and
Social Media**



Data Storage



Insurance



Provenance



IoT



Voting



Supply Chain



Medical Records



Charities

